SHIVAM CHAND KAUSHIK

AI/ML Engineer | Semiconductor Domain Expert

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Professional Summary

AI/ML engineer with 9 years of semiconductor and systems software experience spanning advanced packaging, IC inspection, and factory automation at **Kulicke & Soffa** and **KLA**. Currently pursuing **M.Tech in Artificial Intelligence from IIT Jodhpur**. Building production-grade AI systems—multi-agent architectures, vision-language models, and knowledge graph platforms—specifically for semiconductor manufacturing, inspection, and EDA workflows. One of a rare cohort who combines deep fab-floor domain knowledge (SECS/GEM, wire bonding, pick-and-place, machine vision) with modern AI/ML engineering (LangGraph, PyTorch, LLaVA, RAG, multi-agent orchestration). Proven ability to ship production C++ software across three countries, lead engineering teams, and deliver systems deployed to military and semiconductor customers.

Core Competencies

AI/ML & Deep Learning	Semiconductor Domain	Software & Infrastructure
PyTorch, TensorFlow, torchvision	Wafer Defect Inspection (KLA)	C++ (9 years production)
LangGraph, LangChain, RAG	Adv. Packaging & Wire Bonding (K&S)	Python, C#/.NET, WPF
LLaVA, Qwen-VL, DeepSeek-Coder	SECS/GEM Protocol (Intel integration)	FastAPI, Docker, Kubernetes
XGBoost, scikit-learn, SHAP	Pick & Place (iFlex, Luminex, Pixalux)	MQTT, TCP/IP, WebSocket
Computer Vision, CNNs, ViT	SPC, Yield Analysis, Pred. Maintenance	Neo4j, PostgreSQL, ChromaDB
Multi-Agent Orchestration	Smart Factory / Industry 4.0	React, Streamlit, Git/CI-CD

Work Experience

Software Engineer — Semiconductor Assembly & Smart Factory AI
Kulicke & Soffa (K&S), Netherlands & Singapore — Sept 2020 — Present
via ASM Technologies | K&S is a global leader in semiconductor assembly equipment for advanced packaging.

- Developing and optimizing the C++ software stack for RapidPro and ATPP wire bonding machines, engineering real-time machine control systems that process high-frequency sensor data and manage process parameter optimization.
- Architected product decoupling in the Machine Configuration Tool (MCT), enabling multi-platform support across 6 machine variants (iFlex-T2/T4/H1/CX, Luminex, Pixalux).
- Integrated **MQTT protocol** into MCT for the Smart Factory initiative, enabling real-time machine-to-cloud data streaming for production analytics.
- Led code quality modernization: compiler warning upgrades (W3 to W4), static analysis (PVS-Studio), IWYU implementation, and Boost-to-STL migration.

Software Engineer — IC Inspection & Machine Vision
Quest Global Technologies (client: KLA/ICOS) — July 2018 — Sept 2020
Bangalore | KLA is the world's leading semiconductor process control and inspection equipment company.

- Built a Python-based hardware simulator for the ICOS T800 Gantry System—a TCP/IP server modeling machine behavior for offline testing and digital-twin development.
- Developed data logging and analytics pipelines for production metrics: throughput tracking, taper parameter monitoring, and per-head performance analysis.
- Implemented **SECS/GEM interface for Intel**—the semiconductor industry standard protocol for equipment-to-host communication in automated fabs.

- Engineered granular process control: independent per-head yield-loss diagnostics for YZ1/YZ2/YZ4 pick & place heads.

Software Engineer — Telecom & Embedded Systems
Quest Global Technologies (client: Reliance Jio) 2018
Bangalore

- Developed LibWebSocket server for IoT smart speaker integration with ONT router—embedded systems communication architecture for Jio's IoT ecosystem.

Software Engineer — Aerospace & Defense Systems
AxisCades Aerospace & Technology May 2016 — June 2018
Bangalore

- Developed real-time 2D signal visualization for a Bird Detection Radar system using C++/Qt.
- **Led a team of 5 engineers** and commissioned ACRT across **37 units of the Indian Army and Indian Air Force**.

AI/ML Projects

LLM & Agentic AI Systems

1. Multi-Modal Vision-Language AI for Semiconductor Inspection & Autonomous FA Report Generation

- Built a multi-modal Vision-Language system for semiconductor defect inspection by fine-tuning LLaVA-1.6 with QLoRA on semiconductor image-text pairs, enabling natural language defect description, root cause hypothesis generation, and severity reasoning directly from inspection images (SEM, optical, wafer map).
- Developed four agentic modules (Defect Describer, Root Cause Analyzer, Severity Classifier, Recipe Advisor) that correlate defect images with SECS/GEM equipment logs and a historical defect vector database (Qdrant) to generate structured Failure Analysis reports in under 2 minutes—replacing a 2–4 hour manual process.
- **Stack:** LLaVA-1.6, DINOv2, QLoRA/PEFT, LangGraph, SECS/GEM, MQTT, Qdrant, MinIO, TimescaleDB, FastAPI, Docker

2. Agentic AI for HDL Code Analysis, Generation & Verification

- Architected a multi-agent system for semiconductor HDL verification: four specialized agents (RTL Analyzer, Testbench Generator, Lint/DRC, Design Space Explorer) autonomously parse Verilog/SystemVerilog, generate UVM testbenches with self-correction loops, and optimize design parameters via reinforcement learning—targeting 60–70% of the chip development cycle spent on verification.
- Integrated open-source EDA tools (Icarus Verilog, Yosys, Verilator) as agent tools with RAG over IEEE Verilog standards and design pattern libraries. Supports fully air-gapped deployment with local LLMs for semiconductor IP protection.
- **Stack:** LangGraph, DeepSeek-Coder, pyverilog, Icarus Verilog, Yosys, Qdrant, Stable-Baselines3, Optuna, FastAPI, React, Docker

3. Multi-Agent Autonomous Fab Operations System

- Developed a multi-agent AI system for autonomous semiconductor fab diagnostics. Four specialized agents (Equipment Health, Yield Analyst, Recipe Optimization, Defect Inspector) collaborate through a LangGraph orchestrator to investigate yield excursions end-to-end—from SECS/GEM log anomaly detection to wafer map classification to root cause reporting.
- Implemented real-time MQTT ingestion for live equipment data, RAG over equipment manuals, human-in-the-loop approval gates, and full audit trail for regulated fab compliance. Multi-modal reasoning across SECS/GEM logs, SPC data, and wafer map images.
- **Stack:** LangGraph, PyTorch, MQTT, SECS/GEM, ChromaDB, PostgreSQL, Redis, FastAPI, React, Docker Compose, Prometheus/Grafana

4. LLM-Powered SECS/GEM Protocol Assistant (RAG)

- Built a RAG-powered LLM assistant for the SECS/GEM semiconductor protocol, enabling natural language queries over SEMI standards, equipment specifications, and alarm codes. Fine-tuned retrieval on SECS/GEM documentation with FAISS vector store.
- **Stack:** HuggingFace Transformers, LangChain, FAISS, Streamlit

Deep Learning & Computer Vision

5. Wafer Defect Detection & Classification (WM-811K)

- Built a deep learning pipeline for wafer defect pattern classification on the WM-811K dataset (811K+ wafer maps, 9 defect classes) using ResNet and EfficientNet architectures, with AutoEncoder-based anomaly detection for unseen defect patterns. Achieved 95%+ classification accuracy.
- **Stack:** PyTorch, torchvision, scikit-learn, FastAPI, Docker

Data Science & Manufacturing Analytics

6. Semiconductor Yield Prediction & Explainability (SECOM)

- Developed a yield prediction model on the SECOM semiconductor manufacturing dataset (591 sensor features, 93.4% class imbalance). Identified top 15 critical sensor parameters using SHAP analysis, enabling targeted process monitoring for yield improvement.
- **Stack:** scikit-learn, XGBoost, LightGBM, SHAP, pandas, Streamlit

7. SECS/GEM Data Analytics & Predictive Maintenance

- Built a SECS/GEM data analytics pipeline for semiconductor equipment predictive maintenance. Applied time-series anomaly detection (Isolation Forest, LSTM) with real-time MQTT integration to predict equipment failures before yield impact.
- **Stack:** PyTorch (LSTM), scikit-learn, paho-mqtt, pandas, FastAPI

Process Optimization

8. Wire Bond Quality Prediction & Process Optimization

- Developed ML-based wire bond quality prediction with Bayesian optimization to identify optimal bonding parameters, reducing predicted defect rate by 40%.
- **Stack:** scikit-learn, Optuna, matplotlib

Education

Indian Institute of Technology (IIT), Jodhpur — June 2024 — Present
Master of Technology (M.Tech) — Artificial Intelligence

JSS Academy of Technical Education, Noida — June 2011 — May 2015
Bachelor of Technology (B.Tech) — Information Technology

Achievements

- **Bright Mind Award** — ASM Technologies Ltd. (2021), recognizing exceptional technical contribution in semiconductor software engineering
- **779th Rank** — International Mathematics Olympiad (IMO), demonstrating strong analytical and quantitative aptitude

Certifications (In Progress)

Certification	Provider
Fundamentals of Deep Learning	NVIDIA Deep Learning Institute (DLI)
AWS Certified Machine Learning Engineer — Associate	Amazon Web Services
Azure AI Engineer Associate (AI-102)	Microsoft

Languages

- English — Professional proficiency
- Hindi — Native

Project repositories available on GitHub upon request.